Module 3: Feature Maps and Channels

1. What is a feature map?

A feature map is the output generated when a filter moves across an image or another feature map.

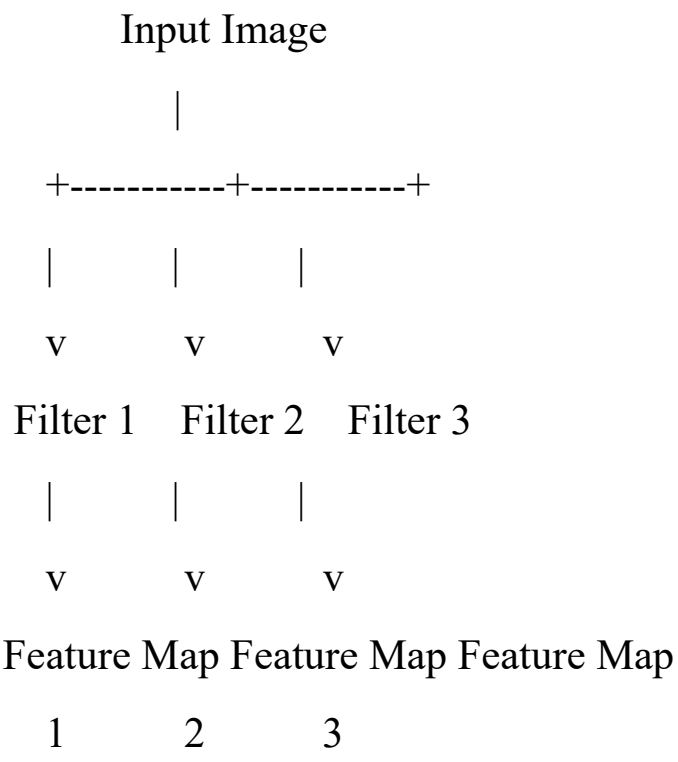
Each value represents the filter's response at a particular location.

For example, a filter that responds to vertical edges may produce stronger values where vertical-edge patterns occur.

2. One filter and multiple filters

A convolutional layer can contain multiple filters.

Each filter learns to respond to different patterns and generates its own output feature map.



If a convolutional layer contains 32 filters, it typically produces 32 output channels.

3. What are channels?

A normal RGB image has three channels:

- Red
- Green
- Blue

After convolution, the output channels are learned feature representations. They are not necessarily colour channels.

For example:

Input image: 128 × 128 × 3

Convolution with 16 filters:

Output: 128 × 128 × 16

This example assumes stride 1 and padding that preserves height and width.

4. How do features become more complex?

Image Pixels

|

v

Edges and Lines

|

v

Textures and Corners

|

v

Shapes and Pattern Combinations

|

v

Complex Visual Representations

This is a simplified explanation. The exact features learned depend on the model and its training data.

5. Feature maps contain numerical values

A feature map is a numerical array. It may be visualized as an image to understand where a network responds strongly, but the values themselves are learned representations.

Conclusion

Filters generate feature maps, and multiple filters produce multiple output channels. This enables CNNs to learn different visual patterns simultaneously.